

### LESSON 324 (1.5 DUAL 1.3 INSTRUMENT)

#### Lesson Objective

During this lesson, the student will be introduced to partial panel and no-gyro radar approaches. With this introduction and a review of attitude instrument flying, the student will obtain the necessary knowledge and skill for the introduction of enroute procedures on the following lesson.

#### Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
- Student questions.

#### Lesson Content (Ground)

##### Go/no-go Decision

- The student must make a go/no-go decision depending on:
  - Personal minimums.
  - FIT minimums.
  - Aircraft.
  - Airspace.
- Review the students planning with a focus on:
  - Weather, weight and balance, performance, flight planning, and fuel calculations.
  - NOTAMs, TFRs, FIFs, aircraft documents, and personal documents.
  - Equipment requirements (e.g. flashlight, fuel card, drinking water, view-limiting devices, life jackets, fly away kit, survival kit, etc.)
  - FRAT/TEM/safety considerations for the flight.
- While this is represented by one line item prior to departure, the go/ no-go decision is continuous and should be a consideration even once airborne.

#### Lesson Content (Flight)

##### DME Arcs

- If time permits, fly down to KVRB to shoot the VOR 12R @ KVRB. If not, assign the student a DME arc from a radial or radar vector to refresh their memory on the procedures.

##### ILS or Localizer

- Tell the student to expect an ILS or localizer approach.
  - Nearby options include the ILS 09R/LOC BC 27L @ KMLB, the ILS 36 @ KTIK, or the ILS 03/21 @ KCOF.
- If picking up the approach request in the air, make sure you show your student how to contact Orlando approach before making them do it.
  - **Weather:**
  - **Instruments:**

- Set both altimeters.
- Set the correct nav source on the PFD (VLOC/GPS)
- Load the procedure into the GPS. Load each procedure from an IAF transition (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
- If activating vectors to final, advise your student to set the appropriate approach course.
- Radios:
  - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
  - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
  - Set the final approach course if required.
- Enroute brief
  - Emphasize the importance of this being a brief and not explaining the plate. Students will tend to miss radio calls as they will be talking over approach. Instructors, make sure you are paying attention to the frequency and also listening to your student briefing the plate. Adjust the comm volumes accordingly.

### Missed approach - partial panel

- During one of the missed approaches, fail the PFD (by turning the brightness down) right at the moment the student goes missed.
  - Make sure to only fail the PFD by turning down the brightness, as resetting the circuit breaker will result in needing to come to a stop for AHRS alignment.
- Ensure they maintain positive control of the aircraft.
- Ensure they rely on the standby instruments while briefly troubleshooting the PFD:
  - Brightness
  - Circuit breaker
- Ensure they comply with the steps of the missed approach procedure:
  - Flaps retract at positive rate
  - Inform ATC of the missed approach
- Use the magnetic compass to comply with ATC-assigned headings.

### No-gyro radar vectors

- At some point during the flight, simulate a failure of the gyroscopic system by dimming the PFD or covering the attitude indicator and HSI.
- In this scenario, the student would need to ask for a no-gyro approach from ATC.
- ATC will provide precise instructions on when to turn/climb/descend.
- If time permits, simulate this during one of the segments of an approach.

### GPS approach - partial panel

- Ask ATC to begin a GPS approach from an IAF (for example, NAJBE for the RNAV 09R @ KMLB).

If given a direct to fix instruction, proceed to this fix via the PROC page.

- Ensure the student completes all the steps of the **WIRE** checklist:
  - **Weather**
  - **Instruments:**
    - Set both altimeters.
    - Set the correct nav source on the PFD (VLOC/GPS).
    - Load the procedure into the GPS. Load each procedure from an IAF (if ATC does not specify, choose one that makes the most sense) unless ATC states the approach will begin from radar vectors.
  - **Radios:**
    - Set COMM radios. Ensure the student sets frequencies for approach, tower, ground, and ATIS.
    - Tune/ID Nav radios. Ensure the student sets not only frequency for the navaid being used, but also any other frequencies the approach includes (for example, the MLB VOR as a part of the missed approach instructions on the ILS 09R).
    - Set the final approach course if required.
  - **Enroute brief**
- Point out the G430 prompts to turn just prior to reaching fixes on the approach so as not to overshoot.
  - If autopilot is to be used, engage GPSS mode by pressing NAV twice to ensure this happens automatically.
- Prior to final approach course intercept and when cleared for the approach, ensure airspeed is 90 knots, extend the flaps to 10°, and set the MFD to the engine page.
- After the final approach course intercept, fail the PFD by turning down the brightness or pulling the circuit breaker.
  - NOTE: pulling the circuit breaker means the AHRS will not be operative until the aircraft comes to a stop for final alignment.
  - Ensure the student completes the memory items for PFD failure:
    - Brightness to maximum
    - Simulate resetting the PFD circuit breaker
  - Show the student how to access the first page of the NAV page on the G430 for lateral guidance.
    - If the FPL page is still displayed from initial setup, press the FPL button to exit.
    - Explain the DTK and TRK functions of this page. DTK (Desired Track) is a straight line coming from the runway indicating the final approach course. TRK (current track) indicates the current ground track of the aircraft.
  - Emphasize the need for very small corrections. If the student gets off course, make no more than a 5-10-degree correction and WAIT for re-intercept.
  - Remind the student of their new minimums (LNAV MDA).
- At the FAF, remind the student that setting around 1,800RPM and pitching for 90 knots will

provide a -500fpm descent rate.

- Ensure the student makes altitude callouts at 1,000ft, 500ft, and 100ft above minimums.
- Ensure the student levels off at the MDA. Remind the student of the 91.175(c) criteria required to land. Upon reaching the VDP, have the student go visual and land.

### **Circling approach**

- On any one of the approaches, request a circle to land.
  - If this cannot be completed at KMLB, it can be completed at X26 without talking to ATC. Make sure to make calls to CTAF and circle at pattern altitude if needed.
- Remind the student of the circling radius airspace.
  - The  symbol indicates expanded circling radius. A lack of this symbol indicates standard circling radius.
- Just prior to the specified distance, the student should go visual without being told, and begin the circle.
  - During their approach briefing, the student should include the direction (left/right) that they will turn when they begin the circle so that they don't have to figure that out now.
- Inform ATC if required.
- Hold the circling MDA throughout the circle.
- Once a landing is assured, descend below the circling MDA, extend the flaps and land.

### **Post-Brief**

- Student Self Critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 325
- Assign Homework:
  - Continue reviewing approach procedures
  - Review weather products and information
  - Purchase a low enroute chart

### **Completion Standards**

The student should:

1. Have a basic understanding of the procedures used to perform partial panel and no-gyro radar approaches.
2. Demonstrate proficiency in copying and complying with ATC clearances.